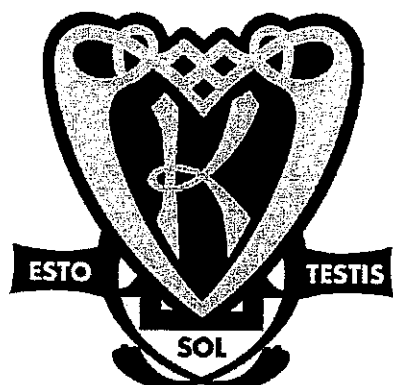


KAMBALA



2013
Higher School Certificate
Trial Examination

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Board approved calculators may be used.
- Write using black or blue pen
- A table of standard integrals is provided at the back of the paper
- All necessary working should be shown in Question 11 – 14
- Write your student number and/or name at the top of every page

Total marks – 70

Section I - Pages 3 – 5

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Section II - Pages 6 – 9

60 marks

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:

Section 1

10 marks

Attempt Questions 1-10

Allow 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

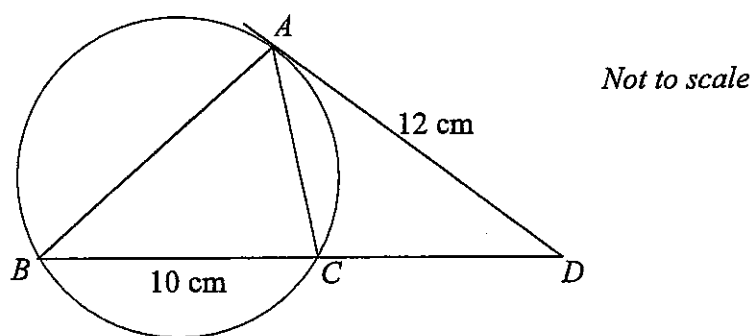
- 1 The remainder obtained when $P(x) = x^3 - 3x^2 - 5x + 6$ is divided by $(x - 3)$ will be: 1

- (A) -34
(B) -9
(C) 6
(D) 21

- 2 Which of the following is an expression for $\frac{d}{dx}(2^x)$? 1

- (A) $x2^{x-1}$
(B) 2^{x-1}
(C) 2^x
(D) $2^x \log_e 2$

3



ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at D where $BC = 10$ cm and $AD = 12$ cm. What is the length of CD ? 1

- (A) 6 cm
(B) 7 cm
(C) 8 cm
(D) 9 cm

Marks

4 The equation $2x^3 + x^2 - 13x + 6 = 0$ has roots α , $\frac{1}{\alpha}$ and β . What is the value of β ? **1**

- (A) 3
- (B) 2
- (C) -3
- (D) -6

5 Which of the following is an expression for $\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right)$? **1**

- (A) $\frac{-x^2}{1+x^2}$
- (B) $\frac{-1}{1+x^2}$
- (C) $\frac{1}{1+x^2}$
- (D) $\frac{x^2}{1+x^2}$

6 Which of the following lines is a horizontal asymptote of the curve $y = \frac{e^x - 2}{e^x + 2}$? **1**

- (A) $y = -2$
- (B) $y = -1$
- (C) $y = 0$
- (D) $y = 2$

7 After t years the number N of individuals in a population is given by $N = 400 + 100e^{-0.1t}$. What is the difference between the initial population size and the limiting population size? **1**

- (A) 100
- (B) 300
- (C) 400
- (D) 500

8 What is the derivative of $y = \cos^{-1}\left(\frac{1}{x}\right)$ with respect to x ?

1

(A) $\frac{-1}{\sqrt{x^2-1}}$

(C) $\frac{1}{\sqrt{x^2-1}}$

(B) $\frac{-1}{x\sqrt{x^2-1}}$

(D) $\frac{1}{x\sqrt{x^2-1}}$

9 Which of the following statements is FALSE.

1

(A) $\cos^{-1}(-\theta) = -\cos^{-1}\theta$

(B) $\sin^{-1}(-\theta) = -\sin^{-1}\theta$

(C) $\tan^{-1}(-\theta) = -\tan^{-1}\theta$

(D) $\cos^{-1}(-\theta) = \pi - \cos^{-1}\theta$

10 Which of the following is the domain of the function $y = \ln\left(x + \sqrt{x^2 + 1}\right)$?

1

(A) $\{x : x \leq -1, x \geq 1\}$

(B) $\{x : -1 \leq x \leq 1\}$

(C) $\{x : x \text{ is an element of the set of real numbers}\}$

(D) $\{x : x \geq 1\}$

Section 2

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Attempt each question in a SEPARATE writing booklet. Extra writing booklets are available. In Questions 11-14 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Use a SEPARATE writing booklet.

- (a) (i) Sketch the function $y = |x^2 - 4|$. 1
- (ii) At what points is $y = |x^2 - 4|$ not differentiable? 1
- (b) $A(-1, 4)$ and $B(7, -2)$ are two points. Find the coordinates of the point P that divides the interval AB internally in the ratio $3 : 2$. 2
- (c) Find correct to the nearest degree the acute angle θ between the lines $3x - 2y = 0$ and $x + 3y = 0$. 2
- (d) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$. 3
- (e) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x} = \cot \frac{x}{2}$. 3
- (f) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$ with focus $F(0, a)$.
- (i) Use differentiation to show that the tangent to the parabola at P has gradient t and equation $tx - y - at^2 = 0$. 2
- (ii) Show that the shortest distance between the focus and this tangent is $a\sqrt{1+t^2}$. 1

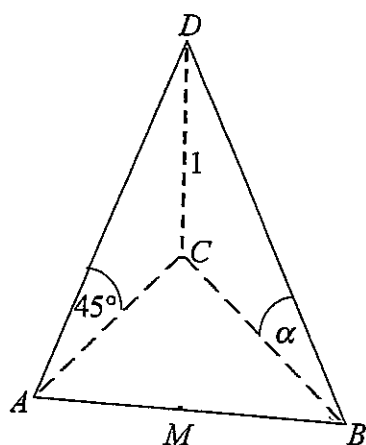
Question 12 (15 marks)

Use a SEPARATE writing booklet.

Marks

- (a) If $x + \frac{1}{x} = 3$, find the value of $x^2 + \frac{1}{x^2}$. 2
- (b) Solve the inequality $\frac{2x-1}{x+2} > 1$. 3
- (c) Use the substitution $u = 6 - x$ to find the exact value of $\int_1^6 x\sqrt{6-x} \, dx$. 3
- (d) Consider the statement:
 $S(n): 2^n - (-1)^n$ is divisible by 3, where n is a positive integer.
- (i) Show that $S(1)$ and $S(2)$ are true. 1
- (ii) Show that if $S(k)$ is true for all positive integers k then $S(k+2)$ is also true. 2

(e)



CD is a vertical pole of height 1 metre that stands with its base C on horizontal ground. A is a point due South of C such that the angle of elevation of D from A is 45° . B is a point due East of C such that the angle of elevation of D from B is α . M is the midpoint of AB .

- (i) Show that $AB = \operatorname{cosec} \alpha$. 2
- (ii) Show that $CM = \frac{1}{2} \operatorname{cosec} \alpha$. 2

Question 13 (15 marks)

Use a SEPARATE writing booklet.

(a) Consider the function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

(i) Show that $f(-x) = -f(x)$

1

(ii) Show that $f(x) = 1 - \frac{2}{e^{2x} + 1}$.

1

(iii) Explain why $f(x) < 1$ for all values of x .

1

(b) Consider the function $f(x) = \sin^{-1}(x-1)$.

(i) Find the domain of the function.

1

(ii) Sketch the graph of the curve $y = f(x)$ showing the endpoints and the x intercept.

2

(iii) The region in the first quadrant bounded by the curve $y = f(x)$ and the y axis between the lines $y = 0$ and $y = \frac{\pi}{2}$ is rotated through one complete revolution about the y axis. Find in simplest exact form the volume of the solid of revolution.

3

(c) A particle is performing Simple Harmonic Motion in a straight line. At time t seconds its displacement from a fixed point O on the line is x metres, given by $x = 4\sqrt{2} \cos\left(\frac{\pi}{4}t - \frac{\pi}{4}\right)$, its velocity is $v \text{ ms}^{-1}$ and its acceleration is $\ddot{x} \text{ ms}^{-2}$.

(i) Find the amplitude and period of the motion.

2

(ii) Find the initial position of the particle and determine if it is initially moving towards or away from O .

2

(iii) Find the distance travelled by the particle in the first 3 seconds of its motion.

2

Marks

Question 14 (15 marks)

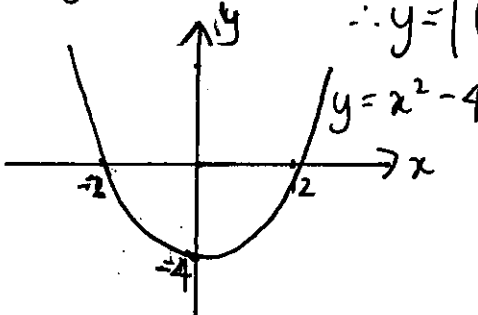
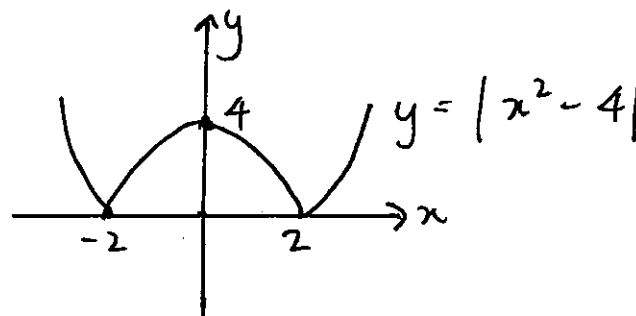
Use a SEPARATE writing booklet.

- (a) Consider the function $f(x) = 2 - \log_e x$.
- (i) Find the equation of the inverse function $f^{-1}(x)$. 1
 - (ii) Explain why the x coordinate X of the point of intersection P of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $e^{2-X} - X = 0$. 2
 - (iii) Use two applications of Newton's Method with an initial value of $X = 1.5$ to find the value of X correct to two decimal places. 3
- (b) A vertical building of height 60 metres stands on horizontal ground. A particle is projected from a point O at the top of the building with speed $V = 20\sqrt{2} \text{ ms}^{-1}$ at an angle α above the horizontal. It moves in a vertical plane under gravity where the acceleration due to gravity is $g = 10 \text{ ms}^{-2}$ and hits the ground at a distance 120 metres from the foot of the building. At time t seconds its horizontal and vertical displacements from O are x metres and y metres respectively, given by $x = 20\sqrt{2} t \cos \alpha$ and $y = 20\sqrt{2} t \sin \alpha - 5t^2$. (Do NOT prove these results.)
- (i) Show that $\alpha = \frac{\pi}{4}$ or $\alpha = \tan^{-1} \frac{1}{3}$. 2
 - (ii) If $\alpha = \tan^{-1} \frac{1}{3}$, find the exact time taken for the particle to hit the ground. 2
 - (iii) If $\alpha = \frac{\pi}{4}$, find the exact speed of the particle after 6 seconds. 2
- (c) The acceleration of a creature is given by $\ddot{x} = -\frac{1}{2}u^2e^{-x}$, where x is the displacement from the origin, and u is the initial velocity at the origin. Given that $u = 2\text{m/sec}$:
- (i) Show that $v^2 = 4e^{-x}$. 2
 - (ii) Explain why $v > 0$. 1

Qn	Solutions	Marks	Comments & Criteria
1.	Section 1 $P(x) = x^3 - 3x^2 - 5x + 6$ remainder = $P(3)$ $P(3) = (3)^3 - 3(3)^2 - 5(3) + 6$ $= -9 \quad \textcircled{B}$		
2.	$\frac{d}{dx} (2^x)$ $= \frac{d}{dx} (e^{\ln 2^x})$ $= \frac{d}{dx} (e^{x \ln 2})$ $= e^{x \ln 2} \cdot \ln 2$ $= 2^x \cdot \ln 2 \quad \textcircled{D}$		
3.	let $CD = x$ $\therefore (x+10)x = (12)^2$ $x^2 + 10x = 144$ $x^2 + 10x - 144 = 0$ $(x+18)(x-8) = 0$ $x = -18, 8$ but x is a length, $\therefore x > 0$ $\therefore x = 8 \text{ cm} \quad \textcircled{C}$		

Qn	Solutions	Marks	Comments & Criteria
4.	$2x^3 + x^2 - 13x + 6 = 0$ roots are $\alpha, \frac{1}{\alpha}, \beta$ $\therefore (\alpha)(\frac{1}{\alpha})(\beta) = -\frac{6}{2}$ $\therefore \beta = -3 \quad \textcircled{C}$		
5.	$\frac{d}{dx} \tan^{-1} \left(\frac{1}{x} \right)$ $= \frac{1}{1 + \left(\frac{1}{x} \right)^2} \left(-\frac{1}{x^2} \right)$ $= \frac{-1}{x^2 + 1} \quad \textcircled{B}$		
6.	$y = \frac{e^x - 2}{e^x + 2}$ $\lim_{x \rightarrow -\infty} \frac{e^x - 2}{e^x + 2}$ $= \frac{0 - 2}{0 + 2}$ $= -1$ $\therefore y = -1 \quad \textcircled{B}$		

Qn	Solutions	Marks	Comments & Criteria
7.	$N = 400 + 100e^{-0.1t}$ at $t=0$, $N = 400 + 100e^0$ $= 400 + 100$ $= 500$ $\therefore$ initial population is 500 $\lim_{t \rightarrow \infty} N = 400$ $\therefore$ difference = $500 - 400$ $= 100 \text{ (A)}$		
8.	$\frac{d}{dx} \cos^{-1}\left(\frac{1}{x}\right)$ $= \frac{-1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \cdot -x^{-2}$ $= \frac{+1}{x^2 \sqrt{\frac{x^2 - 1}{x^2}}}$ $= \frac{1}{x\sqrt{x^2 - 1}} \text{ (D)}$		
9.	(A) (since $\cos^{-1}(-\theta) = \cos^{-1}(\theta)$)		
10.	(C)		

Qn	Solutions	Marks	Comments & Criteria
11a)	Section 2. i) $y = x^2 - 4$ $\therefore y = (x+2)(x-2)$   (ii) not differentiable at $x = \pm 2$ b) A(-1, 4) B(7, -2) $m:n = 3:2$ $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$ $= \left(\frac{(3)(7) + (2)(-1)}{3+2}, \frac{(3)(-2) + (2)(4)}{3+2} \right)$ $= \left(\frac{21-2}{5}, \frac{-6+8}{5} \right)$		

$$P = \left(\frac{19}{5}, \frac{2}{5} \right)$$

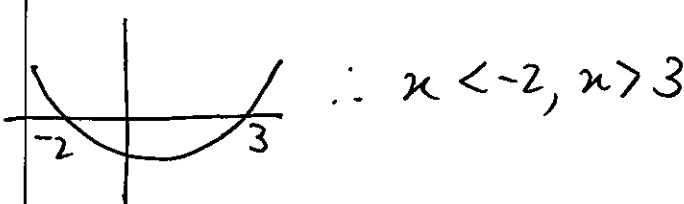
Qn	Solutions	Marks	Comments & Criteria
11c)	$3x - 2y = 0 \quad x + 3y = 0$ $m_1 = \frac{3}{2} \quad m_2 = -\frac{1}{3}$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \frac{\frac{3}{2} + \frac{1}{3}}{1 + (\frac{3}{2})(-\frac{1}{3})}$ $= \frac{\frac{11}{6}}{1 - \frac{1}{2}}$ $= \frac{\frac{11}{6}}{\frac{1}{2}}$ $= \frac{11}{3}$ $\therefore \theta = \tan^{-1} \left(\frac{11}{3} \right)$ $\therefore \theta = 75^\circ \text{ (to nearest degree)}$		

Qn	Solutions	Marks	Comments & Criteria
11d)	$\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ $= \left[\sin^{-1} \frac{x}{2} \right]_{\sqrt{2}}^{\sqrt{3}}$ $= \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1} \left(\frac{\sqrt{2}}{2} \right)$ $= \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$ $= \frac{\pi}{3} - \frac{\pi}{4}$ $= \frac{\pi}{12}$		
11e)	$\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x} = \cot \frac{x}{2}$ let $t = \tan \frac{x}{2}$ $\therefore \text{LHS: } \frac{1 + \cos x + \sin x}{1 - \cos x + \sin x}$ $= \frac{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}$		

Qn	Solutions	Marks	Comments & Criteria
11e) <u>c+d</u>	$= \frac{1+t^2+1-t^2+2t}{1+t^2}$ $\frac{1+t^2-1+t^2+2t}{1+t^2}$ $= \frac{2t+2}{1+t^2}$ $\frac{2t^2+2t}{1+t^2}$ $= \frac{2(t+1)}{1+t^2}, \frac{1+t^2}{2t(t+1)}$ $= \frac{2}{2t}, t^2 \neq -1$ $= \frac{1}{t}$ $= \cot \frac{x}{2}$ $= RHS$		

Qn	Solutions	Marks	Comments & Criteria
11f)	$P(2at, at^2) \quad x^2 = 4ay$ i) $y = \frac{x^2}{4a}$ $y' = \frac{2x}{4a}$ $= \frac{x}{2a}$ at P, $y' = \frac{2at}{2a}$ $= t$ as required eqn: $y - at^2 = t(x - 2at)$ $y - at^2 = xt - 2at^2$ $\therefore tx - y - at^2 = 0$ as required ii) $F(0, a)$ $d = \left \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right$ $= \left \frac{0 + (-1)(a) + (-at^2)}{\sqrt{t^2 + (-1)^2}} \right$ $= \left \frac{-a(1+t^2)}{\sqrt{1+t^2}} \right$		

$$= a\sqrt{1+t^2} \text{ as required}$$

Qn	Solutions	Marks	Comments & Criteria
12a)	$x + \frac{1}{x} = 3$ $\left(x + \frac{1}{x}\right)^2$ $= x^2 + 2 + \frac{1}{x^2}$ $\therefore x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ $= (3)^2 - 2$ $= 7$ b) $\frac{2x-1}{x+2} > 1, x \neq -2$ $(2x-1)(x+2) > (x+2)^2$ $2x^2 + 3x - 2 > x^2 + 4x + 4$ $x^2 - x - 6 > 0$ $(x-3)(x+2) > 0$ 		

Qn	Solutions	Marks	Comments & Criteria
12c)	$\int_1^6 x \sqrt{6-x} \, dx$ $u = 6 - x$ When $x=1$, $u=5$ When $x=6$, $u=0$ $\frac{du}{dx} = -1$ $\therefore dx = -du$ $\int_5^0 (6-u) \cdot \sqrt{u} \cdot -du$ $= \int_0^5 (6u^{\frac{1}{2}} - u^{\frac{3}{2}}) du$ $= \left[\frac{2 \cdot 6u^{\frac{3}{2}}}{3} - \frac{2u^{\frac{5}{2}}}{5} \right]_0^5$ $= \left[4\sqrt{u^3} - \frac{2}{5}\sqrt{u^5} \right]_0^5$ $= \left(4\sqrt{125} - \frac{2}{5}\sqrt{3125} \right) - 0$ $= 20\sqrt{5} - \frac{2}{5} \times 25\sqrt{5}$ $= 20\sqrt{5} - 10\sqrt{5}$ $= 10\sqrt{5}$		

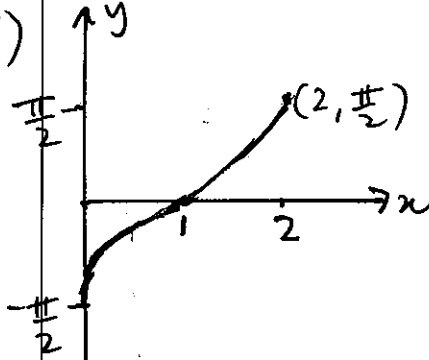
Qn	Solutions	Marks	Comments & Criteria
12d)	$S(n): 2^n - (-1)^n$ divisible by 3 $S(1): 2^1 - (-1)^1$ $= 2 + 1$ $= 3$, divisible by 3 $S(2): 2^2 - (-1)^2$ $= 4 - 1$ $= 3$, divisible by 3 $\therefore S(1), S(2)$ are true ii) If $S(k)$ is true, then $2^k - (-1)^k = 3M$ for some integer M RTP: $S(k+2)$ is true $S(k+2) = 2^{k+2} - (-1)^{k+2}$ $= 2^2 \cdot 2^k - (-1)^k (-1)^2$ $= (3M + (-1)^k) 2^2 - (-1)^k$ $= 12M + 4(-1)^k - (-1)^k$ $= 12M + 3(-1)^k$ $= 3(4M + (-1)^k)$,		

divisible by 3

$\therefore$ true for $S(k+2)$

Qn	Solutions	Marks	Comments & Criteria
12d)	If $S(1)$ and $S(2)$ are true and $S(k+2)$ is true whenever $S(k)$ is true, then, by the Principle of Mathematical Induction, the statement is true for all positive integers n.		
12e)	In $\triangle ABC$: (i) $AC = 1$ $BC = \cot \alpha$ $\angle ACB = 90^\circ$ $\therefore (AB)^2 = 1 + \cot^2 \alpha$ (by Pythagoras' Theorem) $= \operatorname{cosec}^2 \alpha$ $\therefore AB = \operatorname{cosec} \alpha$ as required ii) A unique circle can be drawn through A, B and C. $\angle ACB = 90^\circ$, $\therefore AB$ is a diameter of this circle, with centre M. MC and MB are radii. $\therefore MC = MB = \frac{1}{2} \operatorname{cosec} \alpha$ as required.		

Qn	Solutions	Marks	Comments & Criteria
13a)	$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ i) $f(-x) = \frac{e^{-x} - e^x}{e^{-x} + e^x}$ $-f(x) = - \frac{(e^x - e^{-x})}{e^x + e^{-x}}$ $= \frac{e^{-x} - e^x}{e^{-x} + e^x} = f(-x)$ $\therefore f(-x) = -f(x)$ as required ii) RTP: $f(x) = 1 - \frac{2}{e^{2x} + 1}$ LHS: $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ $= \frac{e^{2x} - 1}{e^{2x} + 1} \quad (\text{multiply by } \frac{e^x}{e^x})$ $= \frac{e^{2x} + 1 - 2}{e^{2x} + 1}$ $= 1 - \frac{2}{e^{2x} + 1}$ $= \text{RHS}$		

Qn	Solutions	Marks	Comments & Criteria
13a) iii)	$\frac{2}{e^{2x}+1}$ is positive for all values of x $\therefore f(x) = 1 - \frac{2}{e^{2x}+1}$ will have a maximum value of 1 $\therefore f(x) < 1$		
13b)	$f(x) = \sin^{-1}(x-1)$ i) Domain: $\{x: -1 < x-1 < 1\}$ $\therefore \{x: 0 < x < 2\}$		
ii)			
iii)	$V = \pi \int_0^{\frac{\pi}{2}} (1 + \sin y)^2 dy$ $= \pi \int_0^{\frac{\pi}{2}} \left\{ 1 + 2\sin y + \frac{1}{2}(1 - \cos 2y) \right\} dy$ $= \pi \left[\frac{1}{2}y - 2\cos y - \frac{1}{4}\sin 2y \right]_0^{\frac{\pi}{2}}$ $= \pi \left\{ \left(\frac{3\pi}{4} - 0 - 0 \right) - (0 - 2 - 0) \right\}$		

$$= \pi \left(\frac{3\pi}{4} + 2 \right) \text{ units}^3$$

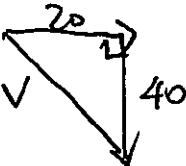
Qn	Solutions	Marks	Comments & Criteria
13c)	$x = 4\sqrt{2} \cos\left(\frac{\pi}{4}t - \frac{\pi}{4}\right)$ i) amplitude = $4\sqrt{2}$ metres period = $\frac{2\pi}{\frac{\pi}{4}}$ $= 8$ seconds ii) at $t = 0$, $x = 4\sqrt{2} \cos\left(0 - \frac{\pi}{4}\right)$ $x = (4\sqrt{2})\left(\frac{1}{\sqrt{2}}\right)$ $x = 4$ m to the right of 0. $\dot{x} = \frac{dx}{dt} = -4\sqrt{2} \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t - \frac{\pi}{4}\right)$ $= -\pi\sqrt{2} \sin\left(\frac{\pi}{4}t - \frac{\pi}{4}\right)$ at $t = 0$, $\dot{x} = -\pi\sqrt{2} \sin\left(0 - \frac{\pi}{4}\right)$ $\therefore \dot{x} = -\pi\sqrt{2} \cdot -\frac{1}{\sqrt{2}}$ $= \pi > 0$ $\therefore$ particle moving away from 0.		

Qn	Solutions	Marks	Comments & Criteria
13c) iii)	at $t=0$, $x=4$ at $t=1$, $x=4\sqrt{2}\cos\left(\frac{\pi}{4}-\frac{\pi}{4}\right)$ $\therefore x=4\sqrt{2}\cos(0)$ $x=4\sqrt{2}$ at $t=2$, $x=4\sqrt{2}\cos\left(\frac{\pi}{2}-\frac{\pi}{4}\right)$ $\therefore x=4\sqrt{2}\cos\left(\frac{\pi}{4}\right)$ $x=4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)$ $\therefore x=4$ at $t=3$, $x=4\sqrt{2}\cos\left(\frac{3\pi}{4}-\frac{\pi}{4}\right)$ $\therefore x=4\sqrt{2}\cos\left(\frac{\pi}{2}\right)$ $x=0$ $\therefore$ Distance travelled $= (4\sqrt{2}-4) + (4-4\sqrt{2}) + (0-4)$ $= 4\sqrt{2}-4-4+4\sqrt{2}+4$ $= (8\sqrt{2}-4) \text{ metres}$		

Qn	Solutions	Marks	Comments & Criteria
14a)	$f(x) = 2 - \log_e x$ i) $\therefore y = 2 - \log_e x$ $\therefore \log_e x = 2 - y$ $x = e^{2-y}$ $\therefore f^{-1}(x) = e^{2-x}$ ii) The graphs of $f(x)$ and $f^{-1}(x)$ are reflections in the line $y = x$, so any points of intersection of these graphs must lie on the line $y = x$. If the graphs intersect at a point where $x = X$, then $f(X) = f^{-1}(X) = X$ $\therefore e^{2-X} = X$ $\therefore e^{2-X} - X = 0$ as required		

Qn	Solutions	Marks	Comments & Criteria
14c) iii)	Let $g(x) = e^{2-x} - x$ $\therefore g'(x) = -e^{2-x} - 1$ $\frac{x - g(x)}{g'(x)} = \frac{x(e^{2-x} + 1) + e^{2-x} - x}{e^{2-x} + 1}$ $= \frac{e^{2-x}(x+1)}{e^{2-x} + 1}$ $= \frac{e^{2-x}(x+1)}{e^{2-x}\left(1 + \frac{1}{e^{2-x}}\right)}$ $= \frac{x+1}{1 + (e^{2-x})^{-1}}$ $= \frac{x+1}{1 + e^{x-2}}$ for $X = 1.5$, $\frac{x - g(x)}{g'(x)} = \frac{2.5}{1 + e^{-0.5}}$ $\doteq 1.56$ for $X = 1.56$, $\frac{x - g(x)}{g'(x)} = \frac{2.56}{1 + e^{-0.44}}$ $\doteq 1.56$ $\therefore X = 1.56$ (to 2 dec. pl.)		

Qn	Solutions	Marks	Comments & Criteria
14b)	$V = 20\sqrt{2}$ $g = 10$ $x = 20\sqrt{2} t \cos \alpha$ $y = 20\sqrt{2} t \sin \alpha - 5t^2$ i) When $x = 120$, $y = -60$ $\therefore t = \frac{x}{20\sqrt{2} \cos \alpha}$ $\therefore 20\sqrt{2} \left(\frac{120}{20\sqrt{2} \cos \alpha} \right) \sin \alpha - 5 \left(\frac{120}{20\sqrt{2} \cos \alpha} \right)^2 = -60$ $120 \tan \alpha - 90 \sec^2 \alpha = -60$ $4 \tan \alpha - 3(1 + \tan^2 \alpha) = -2$ $(3 \tan \alpha - 1)(\tan \alpha - 1) = 0$ $\therefore \tan \alpha = \frac{1}{3}, 1$ $\therefore \alpha = \tan^{-1} \left(\frac{1}{3} \right), \frac{\pi}{4}$ as required ii) $\tan \alpha = \frac{1}{3}$ $\therefore \cos \alpha = \frac{3}{\sqrt{10}}$ $t = \frac{120}{20\sqrt{2} \cos \alpha}$ $\therefore t = \frac{120\sqrt{10}}{20\sqrt{2} \cdot 3}$ $t = 2\sqrt{5}$ seconds		

Qn	Solutions	Marks	Comments & Criteria
14b) iii)	$\alpha = \frac{\pi}{4}$ $\dot{x} = 20 \quad \dot{y} = 20\sqrt{2} \sin\left(\frac{\pi}{4}\right) - 10t$ $\dot{y} = 20\sqrt{2} \cdot \frac{1}{\sqrt{2}} - 10t$ $\dot{y} = 20 - 10t$ at $t = 6$, $\dot{x} = 20$ at $t = 6$, $\dot{y} = 20 - 10(6)$ $\therefore \dot{y} = -40$  $v^2 = 20^2 + 40^2$ $v^2 = 400 + 1600$ $v^2 = 2000$ $v = \sqrt{2000}$ $v = 20\sqrt{5} \text{ ms}^{-1}$		

Qn	Solutions	Marks	Comments & Criteria
14c)	$\ddot{x} = -\frac{1}{2}u^2 e^{-x}$ i) at $t=0$, $u=2$ $\therefore \ddot{x} = -\frac{1}{2}(2)^2 e^{-x}$ $\ddot{x} = -2e^{-x}$ but $\dot{x}' = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ $\therefore \frac{1}{2}v^2 = +2e^{-x} + c$ $v^2 = 4e^{-x} + 2c$ When $t=0$, $x=0$, $v=2$ $\therefore 4 = 4e^0 + 2c$ $\therefore 2c = 0$ $\therefore v^2 = 4e^{-x}$ ii) $4e^{-x} > 0$ for all x $\therefore v^2 > 0$ for all x $\therefore$ at $t=0$, $v=2$ and v remains positive.		